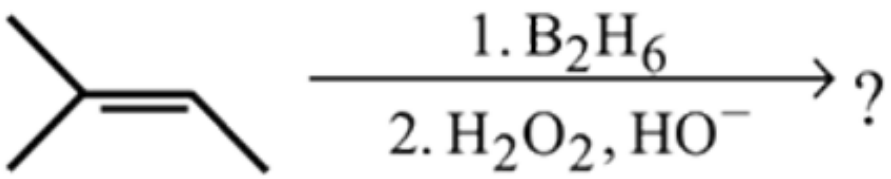
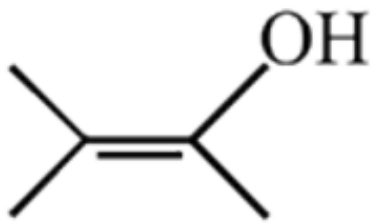
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

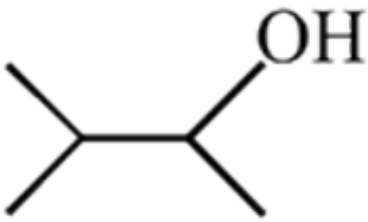
The major product of the following reaction sequence is



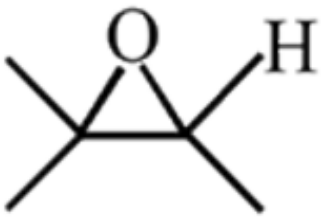
(A)



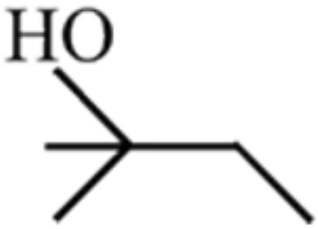
(B)



(C)



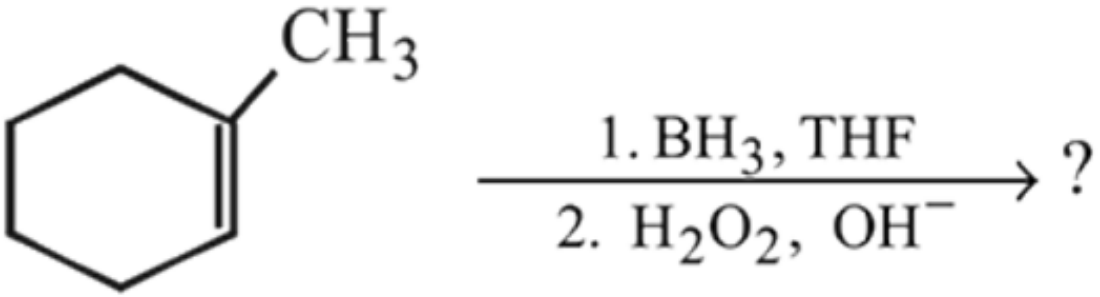
(D)



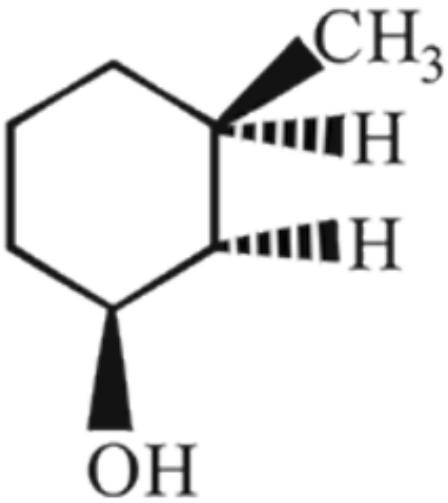
Q.2

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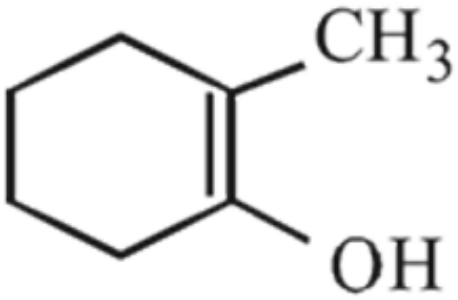
What is the correct product of reaction?



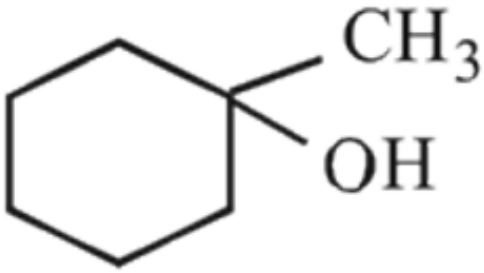
(A)



(B)

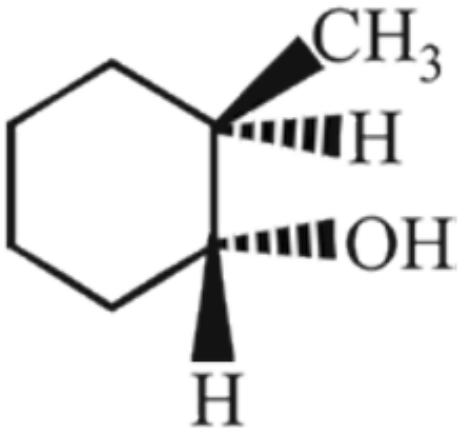


(C)



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(D)



Q.3

In the presence of peroxide, HCl and HI do not give anti-Markownikoff's addition of alkenes because:

(A) One of the steps is endothermic in HCl and HI

(B) Both HCl and HI are strong acids

(C) HCl is oxidizing and the HI is reducing

(D) All the steps are exothermic is HCl and HI

Q.4

Consider the following sequence of reactions

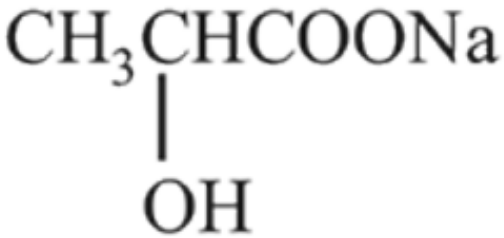
$$CH_3CH=CH_2 \xrightarrow[700\ K]{Cl_2} A \xrightarrow[420\ K, 12\ atm]{Na_2CO_3} B$$

Compound 'B' is

(A)



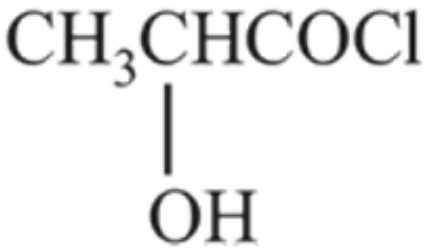
(B)



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(C) $HOCH_2 - CH = CH_2$

(D)



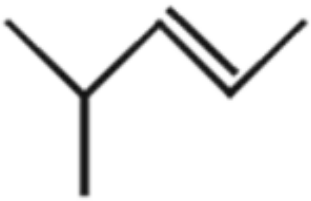
Q.5

Which one of the following compounds gives acetone $(CH_3)_2\,C = O$ as one of the product of its ozonolysis?

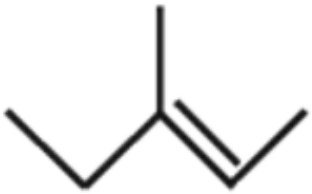
(A)



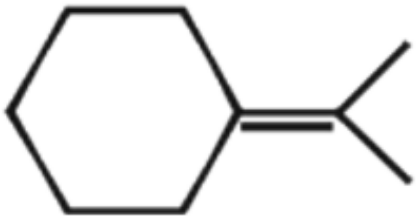
(B)



(C)

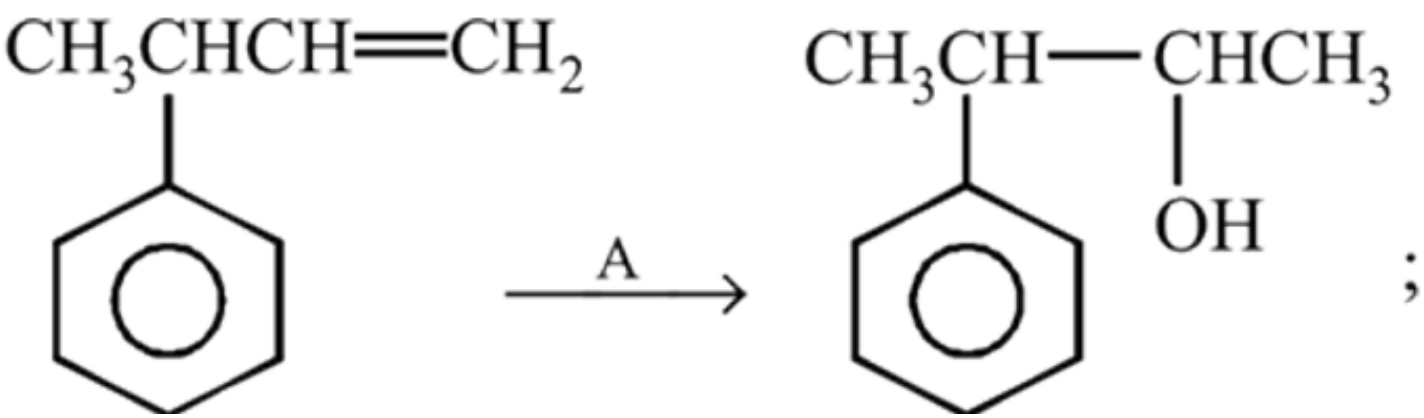


(D)



Q.6

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;

Reagent A may be:

(A) H_2O/H^+

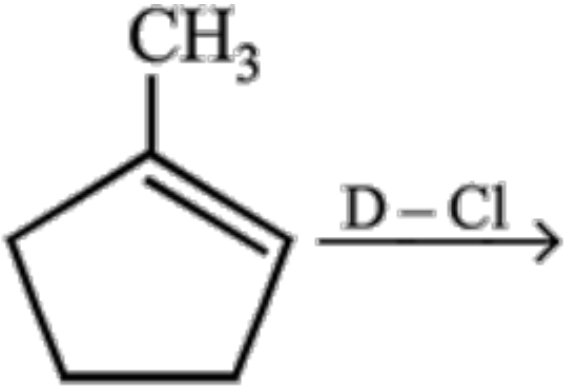
(B) $BH_3, THF/H_2O_2 - OH^-$

(C) $Hg(OCOCH_3)_2, H_2O/NaBH_4 \cdot NaOH$

(D) All are possible

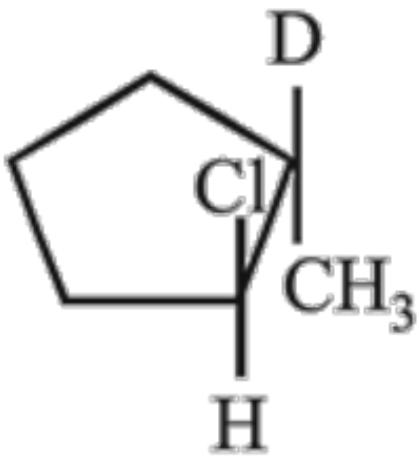
Q.7

What is the major product expected from the following reaction?



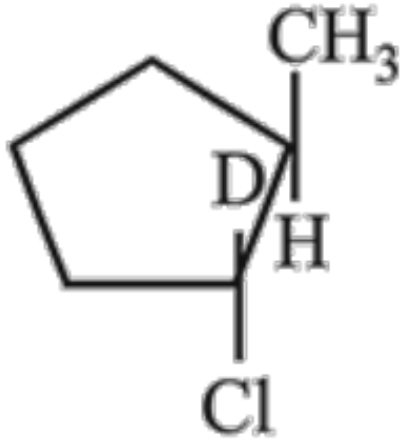
Where D is an isotope of hydrogen

(A)

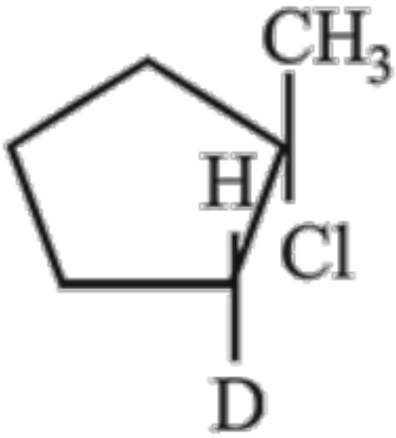


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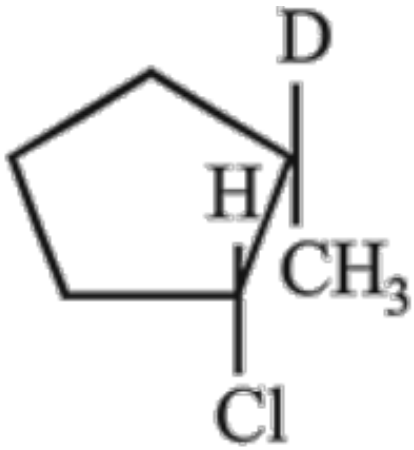
(B)



(C)



(D)



Q.8

Addition of HI to double bond of propene yields isopropyl iodide and not n -propyl iodide as the major product, because addition proceeds through

(A) a more stable carbonium ion

(B) a more stable carbanion

(C) a more stable free radical

(D) homolysis

Q.9

The principal organic product formed in the reaction,

$CH_2 = CH(CH_2)_8COOH + HBr \xrightarrow{\text{peroxide}}$ is

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(A) $CH_3CHBr\,(CH_2)_8\,COOH$

(B) $CH_2 = CH\,(CH_2)_8\,COBr$

(C) $CH_2BrCH_2\,(CH_2)_8\,COOH$

(D) $CH_2 = CH\,(CH_2)_7\,CHBrCOOH$

Q.10

Acid catalyzed hydration of alkenes except ethene leads to the formation of

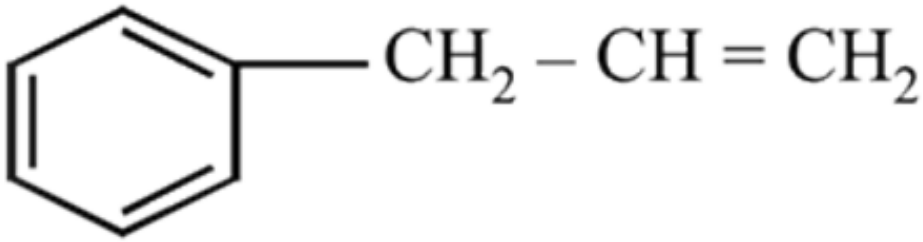
(A) mixture of secondary and tertiary alcohols

(B) mixture of primary and secondary alcohols

(C) secondary or tertiary alcohol

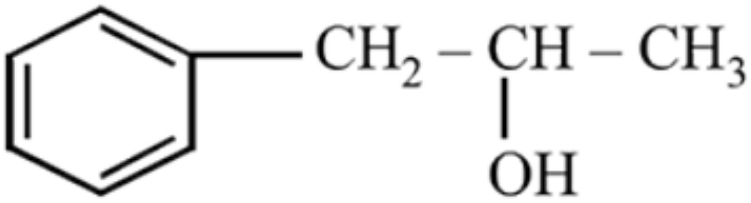
(D) primary alcohol

Q.11

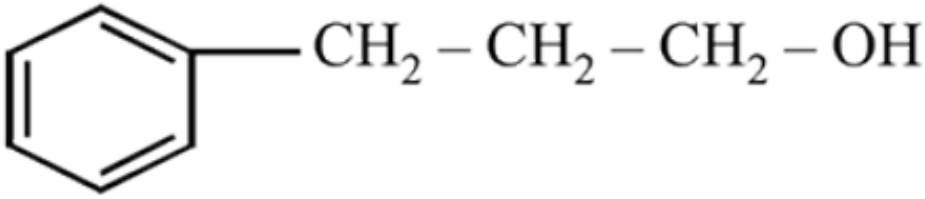


On mercuration-demercuration produces the major product:

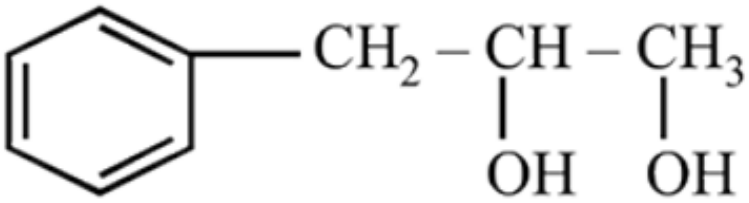
(A)



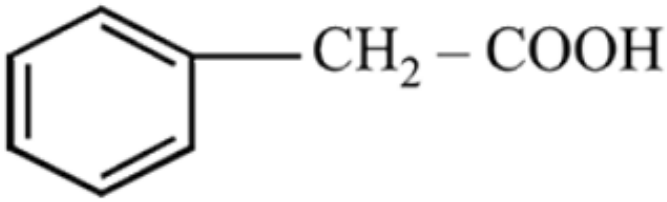
(B)



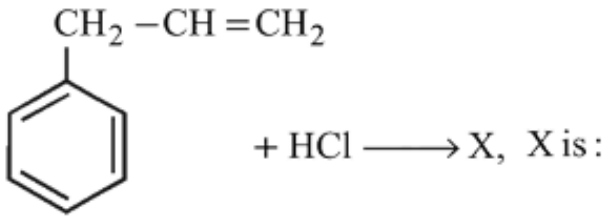
(C)

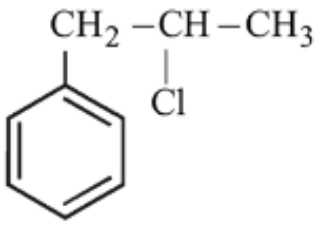


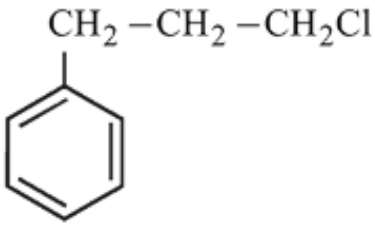
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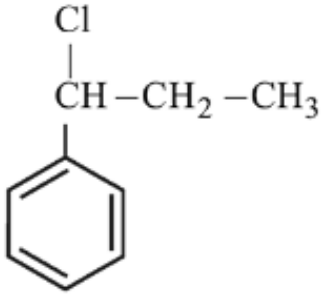
(D) 

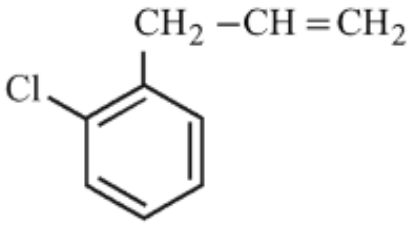
Q.12



(A) 

(B) 

(C) 

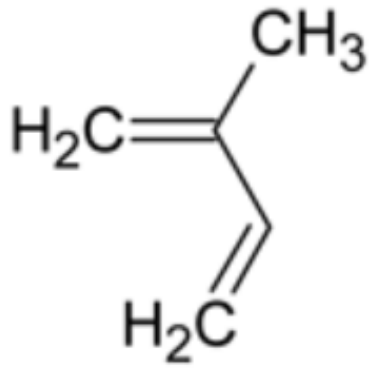
(D) 

Q.13

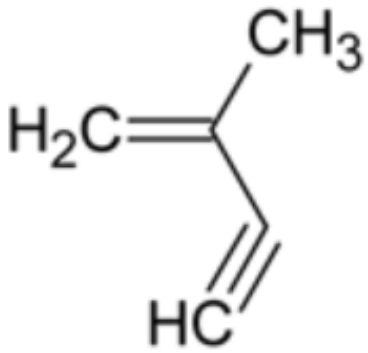
An organic compound $\mathrm{A}\ (\mathrm{C_5H_8})$ on hydrogenation gives compound $\mathrm{B}\ (\mathrm{C_5H_{12}}\ )$. Compound A on reductive ozonolysis give methanal and 2-ketopropanal. The compound (A) is

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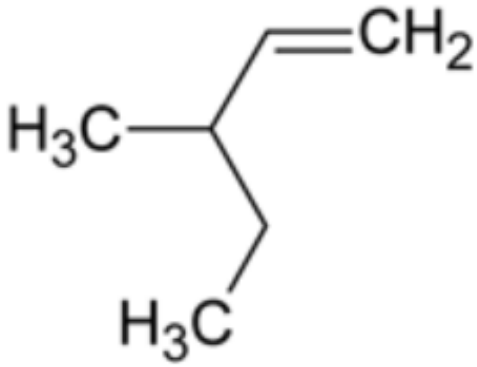
(A)



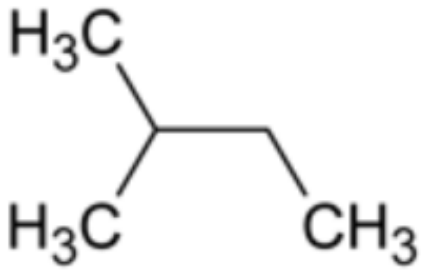
(B)



(C)



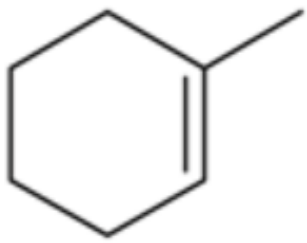
(D)



Q.14

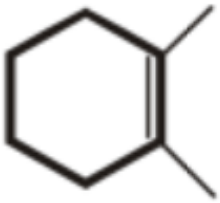
Hydroboration oxidation and acid hydration will not give the same products in case of

(A)



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(B)



(C) $CH_2 = CH_2$

(D) $CH_3CH = CH - CH_3$

Q.15

The eddition of Br_2 to trans-But-2-ene produces

(A) $(+)2,3$ -dibromobutane

(B) $(-)2,3$ -dibromobutane

(C) Racemic-2, 3-dlbromobutane

(D) meso-2, 3-dbromobutane

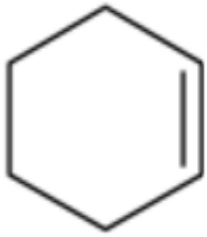
Q.16

In the reaction sequence $\mathrm{A} \xrightarrow{O_3/H_2O} \mathrm{B} \xrightarrow{\Delta} 2CH_3COOH + 2CO_2$ the reactant A is

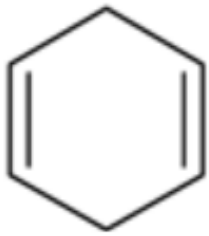
(A)



(B)



(C)

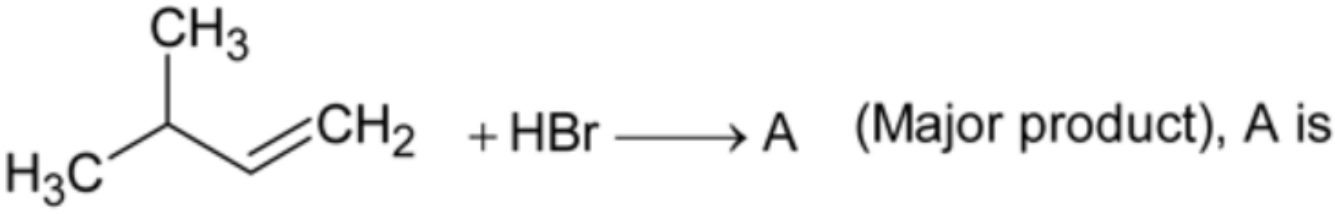


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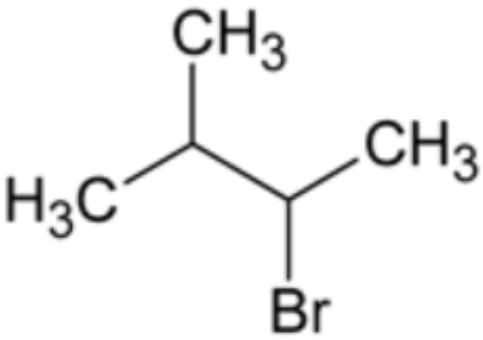
(D)



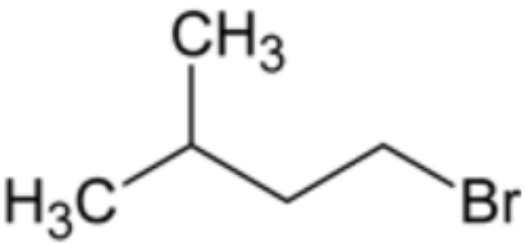
Q.17



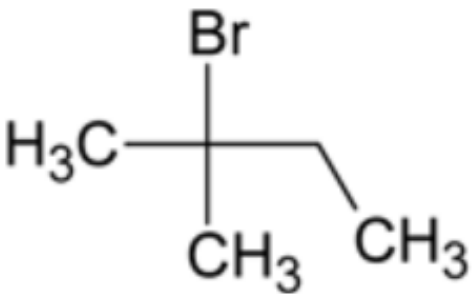
(A)



(B)



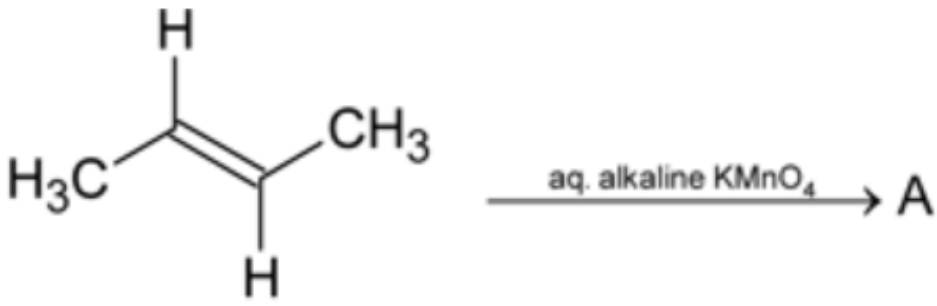
(C)



(D) None is correct

Q.18

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Which of the followng is true about this reaction?

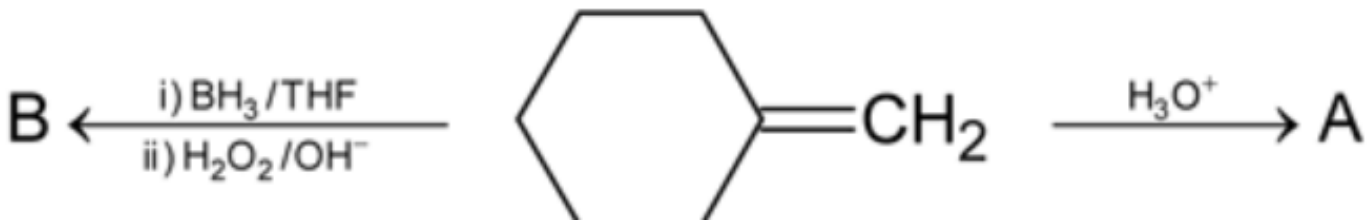
(A) A is meso-1, 2-butanediol formed by syn addition

(B) A ismeso-1, 2-butanediol formed by anti addition

(C) A is a racemic mixture of d and $\mathbf{H}, \mathbf{2}$ -butanediol formed by anti addition.

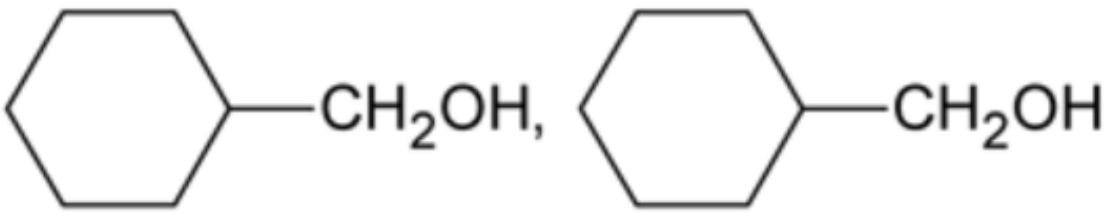
(D) A is a racemix mlxture of d and $1-1,2$ -butanedlol formed by syn addition

Q.19

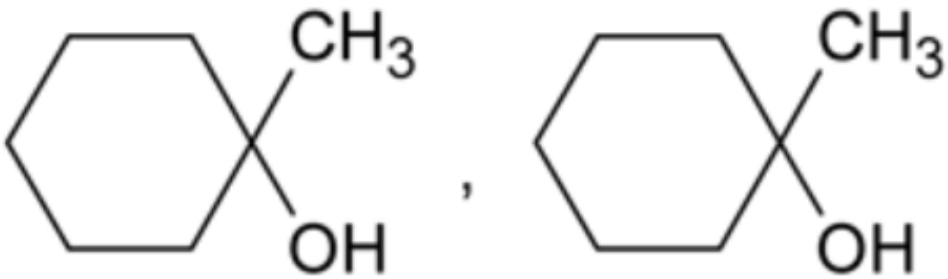


A and B are respectively

(A)

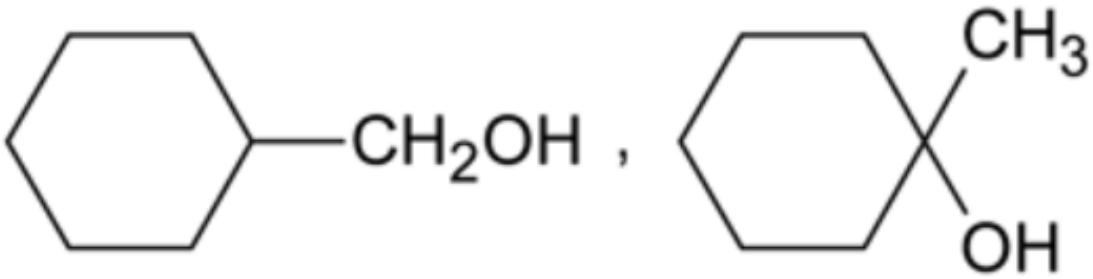


(B)

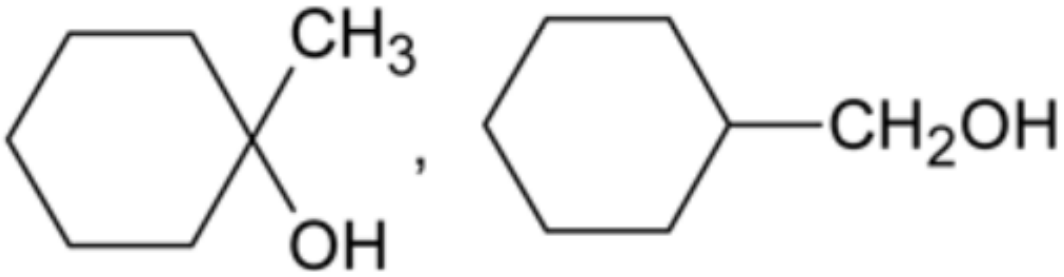


@IITJEE_Advanced

(C)



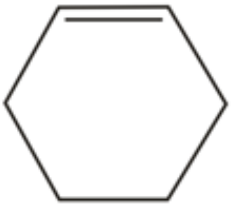
(D)



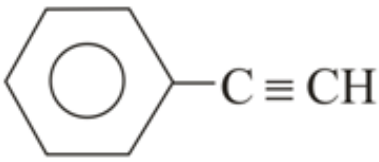
Q.20

Which of the following compound does not decolourise bromine-water solution ?

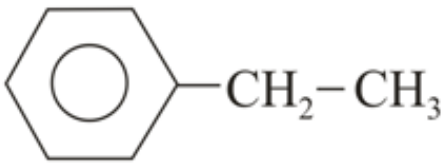
(A)



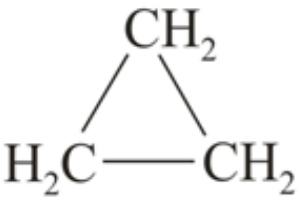
(B)



(C)



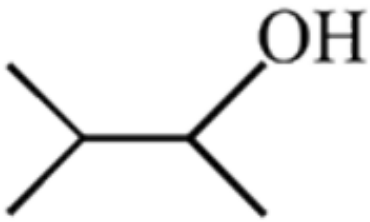
(D)



Answers & Solutions

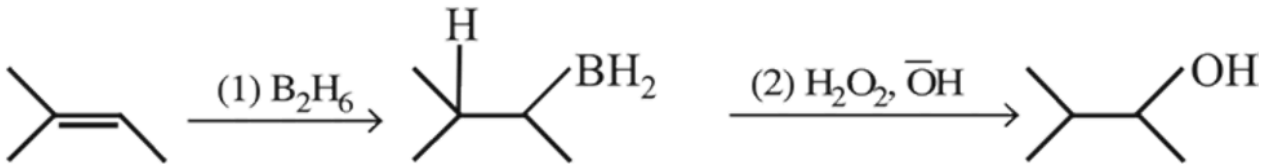
Q.1 Answer:

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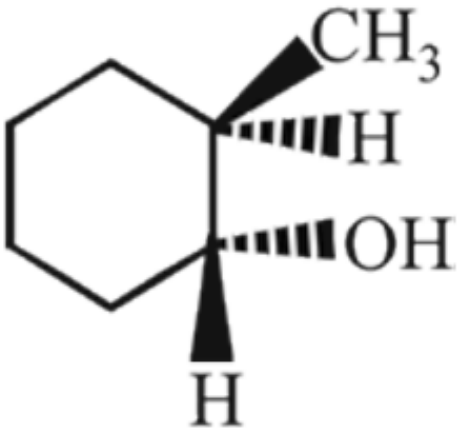


Solution:

Hydroboration oxidation reaction.



Q.2 Answer:



Solution:

The reaction is hydroboration which is addition of H_2O to alkene (anti Makownikoff's's rule)

Q.3 Answer:

One of the steps is endothermic in HCl and HI

Solution:

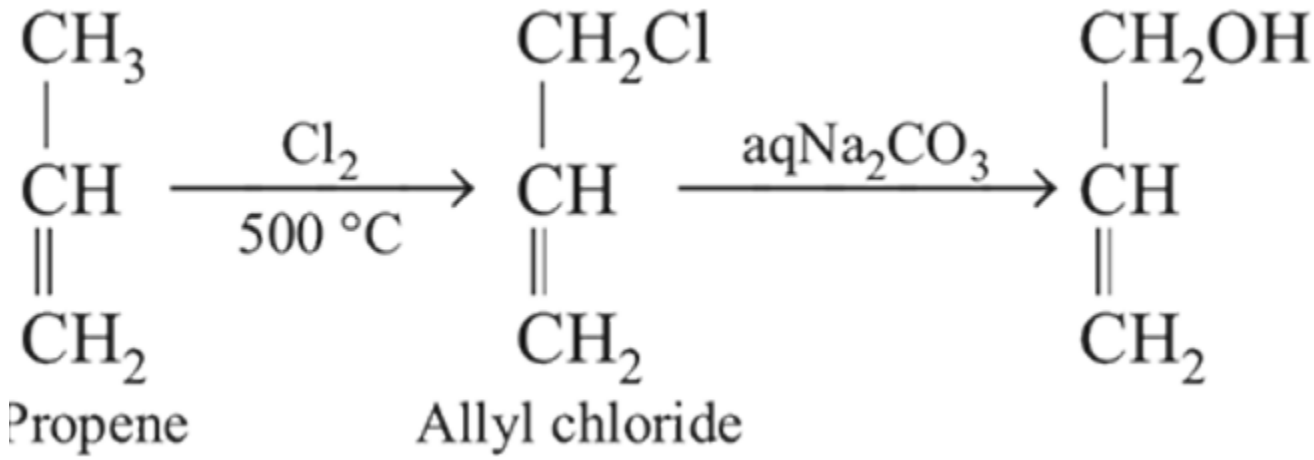
Anti-Markownikoff addition is possible only in case of HBr and not in HCl and HI. In HBr both the chain initiation and propagation steps are exothermic, while in HCl, first step is exothermic, and second step is endothermic and in HI, no step is exothermic. Hence HCl and HI do not undergo antiMarkownikoff's addition.

Q.4 Answer:

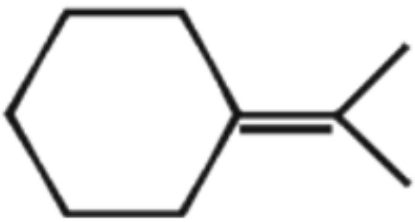
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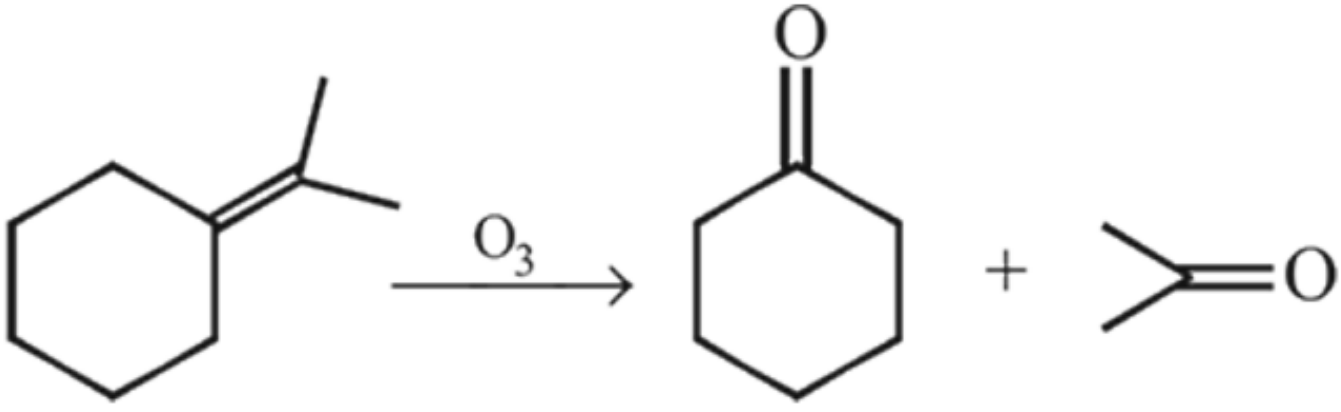
Solution:



Q.5 Answer:



Solution:



Q.6 Answer:

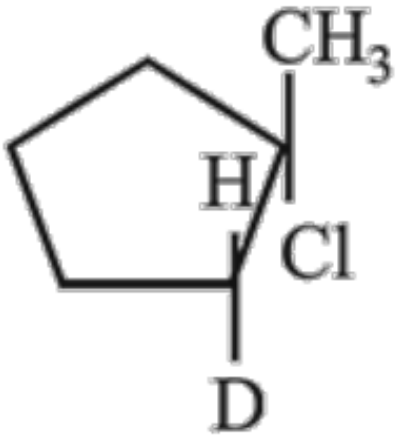
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$Hg(OCOCH_3)_2, H_2O/NaBH_4 \cdot NaOH$

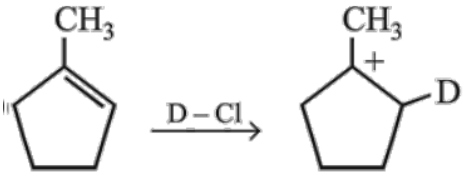
Solution:

Oxymercuration - demercuration reaction addition of H_2O molecule is according to Markovnikoff's rule.

Q.7 Answer:



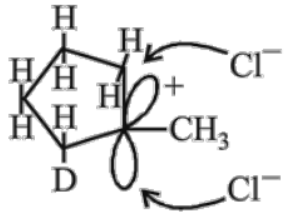
Solution:



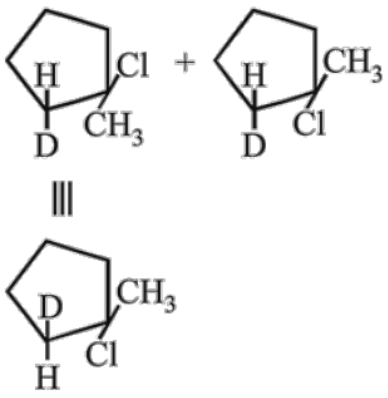
Formation of above species is more favourable. The stability order of carbocations is

$$-\overset{1^\circ}{C}^{+} < -\overset{2^\circ}{\underset{|}{C}}^{+} < -\overset{3^\circ}{\underset{|}{C}}^{+}$$

Now, carbocation formed is sp^2 hybridised that is triangular planar as shown below



Either the $Cl^{\ominus}$ attack from above or below formation of the plane of the molecule, so there is probability of resultant product as shown below i.e., both product will form.



Q.8 Answer:

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a more stable carbonium ion

Solution:

The addition of HI on the double bond of propene yields isopropyl iodide and not n-propyl iodide as the major product. This is because the addition proceeds through a more stable carbonium ion.

$$HI \rightarrow H^+ + I^- \quad CH_3 - CH = CH_2 + H^+ \rightarrow \underset{\substack{\text{(Minor)}\\ 1^\circ \text{ Carbonium ion}\\ \text{Less stable)}}}{CH_3 - CH_2 - \overset{+}{C}H_2} + \underset{\substack{\text{(Major)}\\ 2^\circ \text{ Carbonium ion}\\ \text{(More stable)}}}{CH_3 - \overset{+}{C}H - CH_3}$$

$$\underset{2^\circ \text{ Carbonium ion}}{CH_3 - \overset{+}{C}H - CH_3} + I^- \rightarrow \underset{\substack{\text{Isopropyl iodide}\\ \text{(Major product)}}}{CH_3 - \underset{I}{\underset{|}{CH}} - CH_3}$$

Q.9 Answer:

$CH_2BrCH_2(CH_2)_8COOH$

Solution:

A molecule of HBr is added to $C = C$ double bond. Br is added to less substituted terminal carbon atom. (The anti markonikov's addition of HBr to unsymmetrical alkenes in presence of peroxides)

Q.10 Answer:

secondary or tertiary alcohol

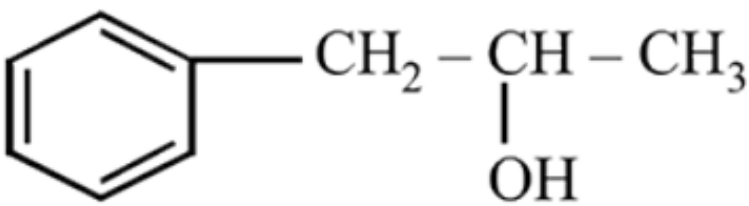
Solution:

$$CH_3 - CH = CH_2 + H_2O \xrightarrow{H_2SO_4} \underset{\substack{\text{OH}\\ 2^\circ \text{ alcohol}}}{CH_3 - \underset{\text{OH}}{\underset{|}{CH}} - CH_3}$$

$$CH_3 - \underset{CH_3}{\underset{|}{C}} = CH_2 + H_2O \xrightarrow{H_2SO_4} \underset{\substack{\text{OH}\\ 3^\circ \text{ alcohol}}}{CH_3 - \underset{\text{OH}}{\underset{|}{\underset{CH_3}{\underset{|}{C}}}} - CH_3}$$

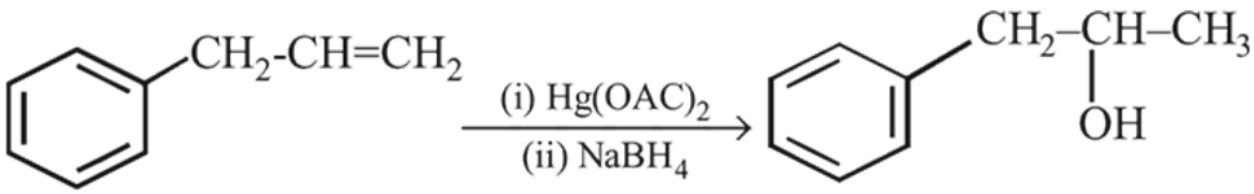
Addition follows Markovnikov's rule.

Q.11 Answer:

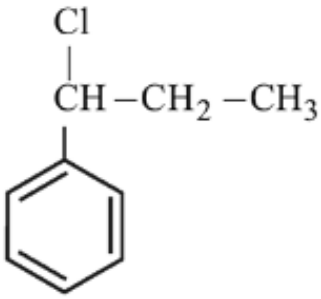


Solution:

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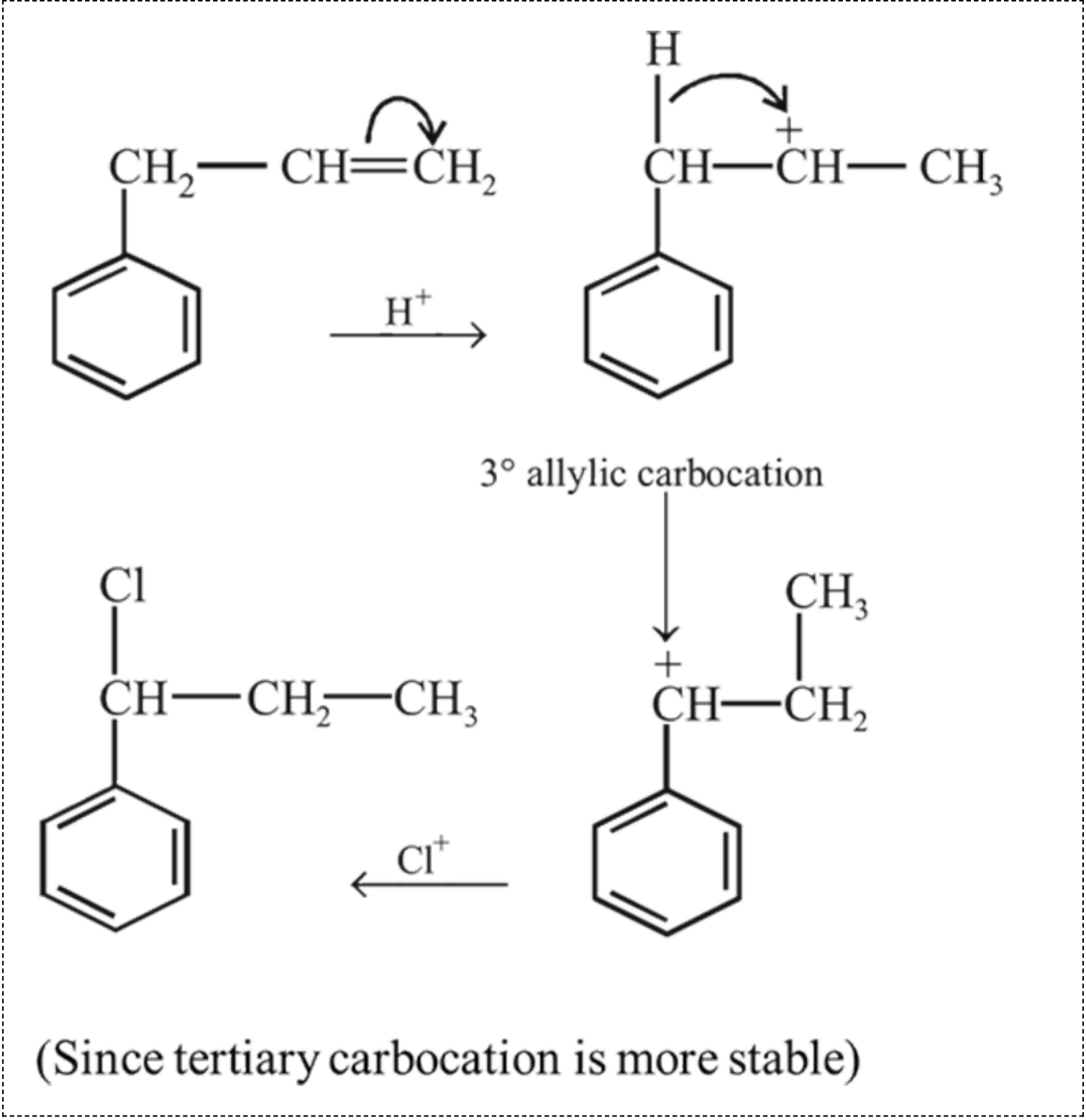


Q.12 Answer:

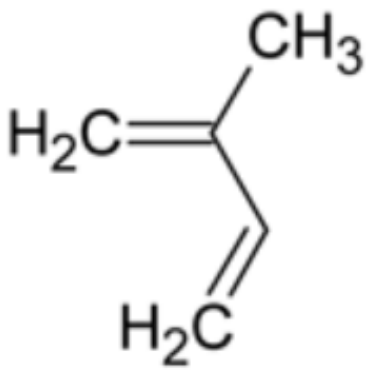


Solution:

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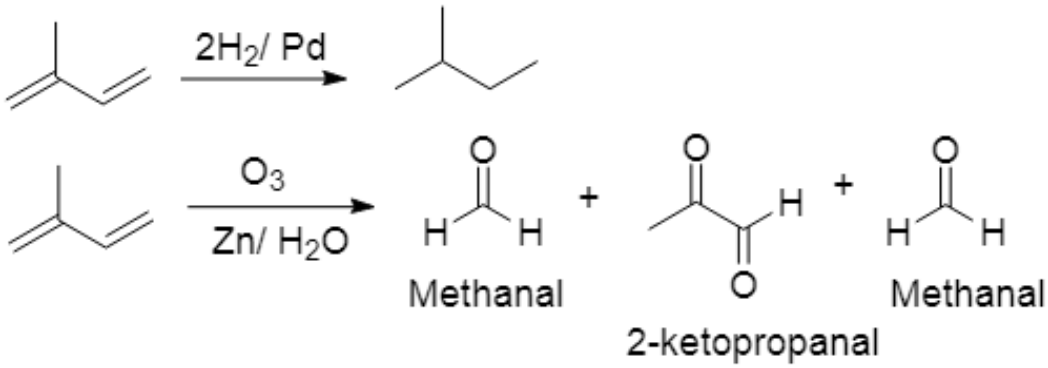


Q.13 Answer:

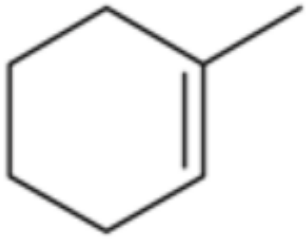


Solution:

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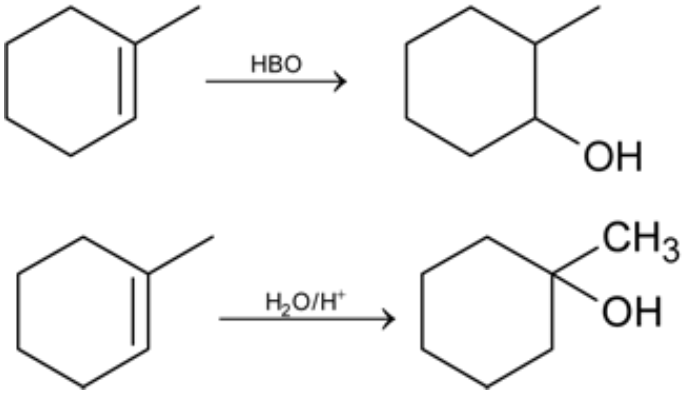


Q.14 Answer:



Solution:

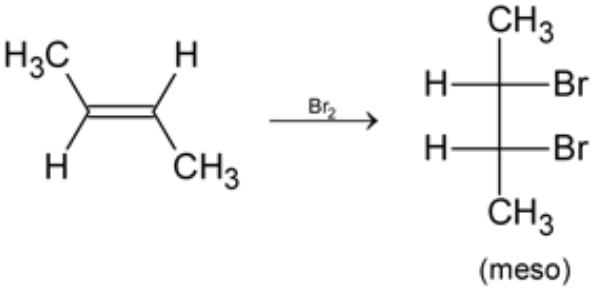
Unsymmetrical alkenes give different products



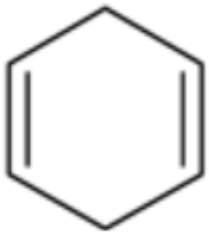
Q.15 Answer:

meso-2, 3-dbromobutane

Solution:

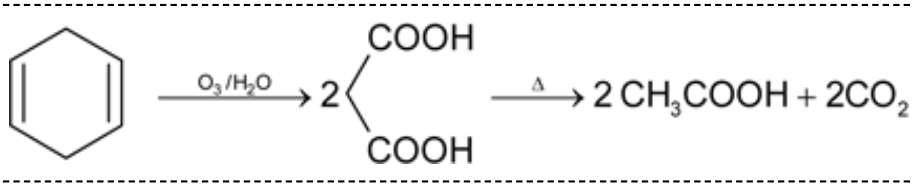


Q.16 Answer:

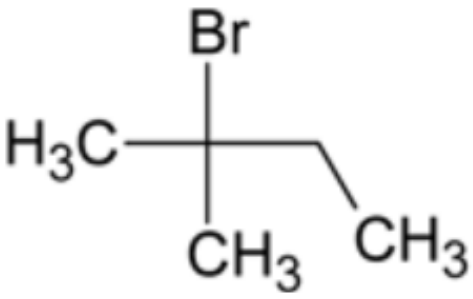


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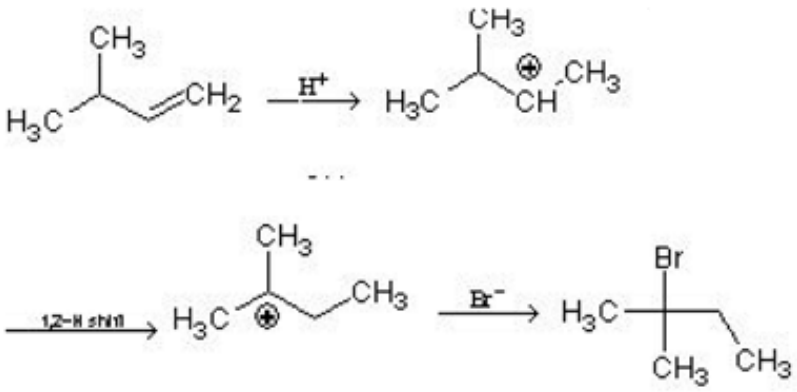
Solution:



Q.17 Answer:



Solution:

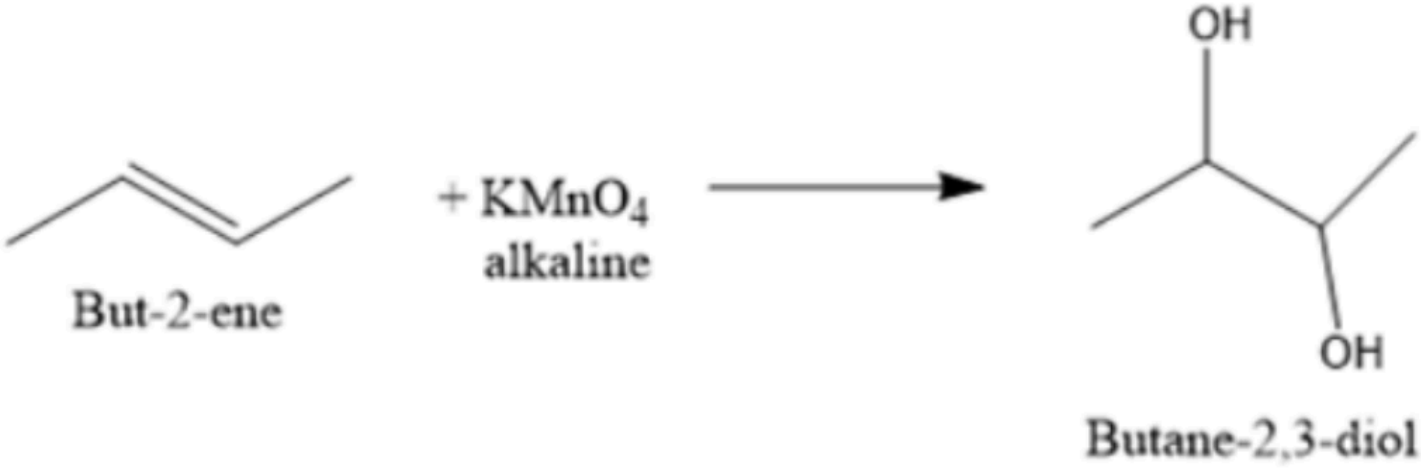


Q.18 Answer:

A is meso-1, 2-butanediol formed by syn addition

Solution:

Alkenes give diol with alkaline $KMnO_4$.



Q.19 Answer:

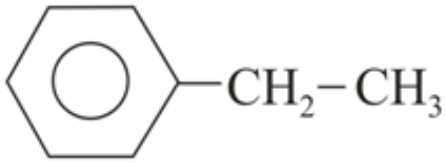
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Solution:

Fomation of A is hydration by Markownikov's orientation. Fomation of B is by HEO reaction (An Markomilkov's orentation)

Q.20 Answer:



Solution:

benzene not decolurise bromine water solution , and only 3 option has 0 degre of unsaturation

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